

4. A large research project was carried out to investigate the effect of the environment on the shell length of common European limpet shells (*Patella vulgata*). These molluscs live on rock surfaces and remain clamped to the rock with a muscular foot when they are exposed at low tide and only move for feeding when they are covered at high tide as shown in **Image 4.1**. They graze algae from the rock surface and always return to the same home position as the tide goes out.

Image 4.1



Patella vulgata clamped to rock

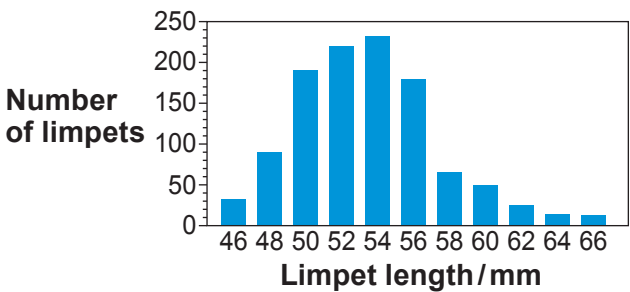


Showing muscular foot

The results from two sites, sheltered and exposed rocky shores, are shown on **Graphs 4.2A** and **4.2B**. The same size area was investigated at both sites.

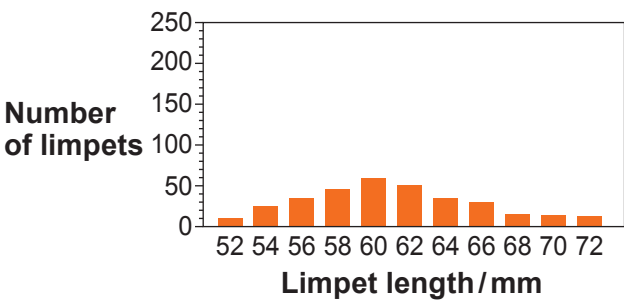
Graph 4.2A

Sheltered rocky shore



Graph 4.2B

Exposed rocky shore



- (a) (i) State **two** conclusions that you could draw from the data in **Graph 4.2A** and **Graph 4.2B**. [2]

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- (ii) State the modes for the two sets of data. [1]

Sheltered rocky shore mode =

Exposed rocky shore mode =



Two groups of A level students carried out the same type of investigation on the two types of rocky shore. They measured the shell length in mm of 20 mature *Patella vulgata* at each site. They recorded their data in a results table and used an on-line calculator which calculated the t value.

Group 1: exposed rocky shore, Cemlyn Bay, Anglesey

Group 2: sheltered rocky shore, Menai Strait

The online results are shown in **Image 4.3**.

Image 4.3

Result

You entered the following data:

Group 1

72 68 59 69 58 58 49 59
61 69 75 80 49 68 70 71
67 58 60 71

Group 2

68 47 45 60 59 49 51 67
58 59 49 67 63 66 61 58
55 60 59 62

Summary

	Group 1	Group 2
Mean	64.55	58.15
Variance	68.05	47.1868
Standard deviation	8.2492	6.8693
n	20	20
t	2.6662	

The t value was calculated as 2.6662

(b) (i) State the null hypothesis for this experiment. [1]

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(ii) State the degrees of freedom which should be used. [1]

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Table 4.4 shows the critical values for a two-tailed test.

Table 4.4

Degrees of freedom	Probability				
	0.1	0.05	0.01	0.005	0.001
30	1.697	2.042	2.75	3.030	3.646
31	1.696	2.040	2.744	3.022	3.633
32	1.693	2.037	2.738	3.015	3.622
33	1.692	2.035	2.733	3.008	3.611
34	1.691	2.032	2.728	3.002	3.601
35	1.690	2.030	2.724	2.996	3.591
36	1.684	2.028	2.719	2.991	3.582
37	1.683	2.026	2.715	2.985	3.574
38	1.682	2.024	2.712	2.980	3.566
39	1.681	2.023	2.708	2.976	3.558
40	1.680	2.021	2.704	2.971	3.551

(iii) Use Table 4.4 to determine whether you would accept or reject the null hypothesis. Explain how you have come to this conclusion.

[3]

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(c) Suggest **one** density-independent and **two** density-dependent factors which could account for the difference in shell length.

[3]

Density-independent factor

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Density-dependent factors

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